

Water Rocket Simulator

Standard

Advanced

Dark Shadow

Charlie

Reset to Default

Rocket

Rocket Outside Diameter (mm)

85

Nozzle Exit Diameter (mm)

7

Total Volume (L)

4.5

Water Amount (L)

1

Empty Mass (g)

803

Drag Coefficient

0.50



Launcher

Pressure (bar)

8

116 psi

Launch Tube Diameter (mm)

0

Launch Tube Length (mm)

0

Run Simulation

Simulation Results

Burnout	
Time	1.81 s
Altitude	31.46 m
Velocity	19.44 m/s
Acceleration	-1.08 G

Apogee	
Time	3.75 s
Altitude	50.00 m (164 feet)

Thrust	
Total Impulse	45.16 Ns
Burn Time	1.42 s
Peak Thrust	60.49 N
Average Thrust	31 N
Motor Equivalent	F31

Crashdown	
Time	6.99 s
Velocity	-29.75 m/s

Graphs

